Oswin Franco Cervantes

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Statement of Purpose

My goal is to use first-principles physics, electromagnetic modeling, circuit and system-level design, measurement, and computational methods to develop integrated hardware spanning semiconductor circuits, RF, photonics, and quantum systems.

Education

Massachusetts Institute of Technology, Cambridge, MA — **Expected May 2028**

B.S. Candidate in Physics and Electrical Engineering with Computing — Sept. 2024–Present

Selected coursework: **Computer Science:** Mathematics for Computer Science (Discrete Mathematics), Computational Structures, Introduction to Low-Level Programming (C/Assembly); **EE/Physics:** Semiconductor Electronic Circuits, Circuits and Electronics, Signal Processing, Vibrations and Waves, Quantum Physics I, Statistical Mechanics

Georgia Tech, Atlanta, GA — Dual Enrollment, Mathematics/Computer Science; GPA: 4.0/4.0 — **Aug. 2022–May 2024**

Integrated Circuit Design Project

22 nm FinFET Amplifier and Control Chip — **Spring 2026**

MIT Semiconductor Electronic Circuits (6.2080) — *Cambridge, MA*

- Designed and simulated a 22 nm FinFET CMOS chip combining an analog amplifier, activation switches, and digital control logic; applied semiconductor device physics and large- and small-signal models.
- Created the chip layout in Cadence for a course tapeout and ran Calibre DRC/LVS checks to verify fabrication rules and consistency with the schematic.
- Verified through simulation that the amplifier exceeded targets of 26 dB gain at 1 MHz and 300 mV peak-to-peak output swing while operating 48% below its 500 μW power limit.

Experience

Undergraduate Researcher — Prof. William D. Oliver — **Sept. 2025–Present**

MIT Research Laboratory of Electronics, Engineering Quantum Systems (EQuS) Group — *Cambridge, MA*

- Serve as technical owner for the on-chip microwave-filter subsystem within a cross-functional quantum-hardware team.
- Design filters for operation from room temperature to cryogenic conditions, applying transmission line and impedance matching principles to suppress out-of-band noise while minimizing passband insertion loss at the interface between semiconducting quantum-dot qubits and superconducting circuits.
- Use PyAEDT to automate 3D model generation and HFSS Driven Terminal and Eigenmode simulations for S-parameter and resonant-mode analysis; use Q3D for quasi-static parasitic extraction and AWR Microwave Office for circuit models.
- Maintain the filter component library in CDES, an object-oriented Python framework; use NumPy, pandas, and gdstk to automate generation of fabrication-ready GDS layouts.
- Fabricate superconducting niobium microwave filters on silicon substrates using photolithography in a cleanroom environment.

Avionics Engineer — **Jan. 2026–Present**

MIT Rocket Team — *Cambridge, MA*

- Design RF circuits and PCB interconnects in Altium; analyze interconnect behavior with HFSS Driven Terminal and Ansys SIwave; compare simulated S-parameters with vector network analyzer measurements.

Hardware Development Engineer (Systems Engineering) Intern — **May 2025–Aug. 2025**

Amazon Lab126, Advanced Products — *Sunnyvale, CA*

- Collaborated with a cross-functional team to translate product requirements into system architecture and functional block diagrams, then designed schematics and PCB layout for a multi-IC solar-charging module for Kindle devices.
- Designed and validated buck, boost, and buck-boost power circuits using OrCAD and SPICE for circuit design and simulation and Cadence Allegro for PCB layout; characterized prototypes with oscilloscopes and electronic loads and investigated maximum power point tracking (MPPT).
- Implemented I^2C communication and assembled SMD prototypes to verify end-to-end circuit functionality; trained new hires on PCB design tools.

Technical Skills

Design and Analysis: Electromagnetic modeling, RF/microwave circuits, analog/mixed-signal ICs, power circuits, S-parameters, transmission lines, impedance matching, SPICE, PCB/system design

Fabrication and Test: 22 nm FinFET custom layout, DRC/LVS, GDS/KLayout, niobium-on-silicon photolithography, cleanroom processing, SMD, VNA and oscilloscope measurements

Software and HDL: Python, C, Java, Julia, MATLAB, Bash, VHDL, Bluespec HDL, RISC-V assembly; NumPy, pandas, gdstk, Git

EDA Tools: Ansys (HFSS, Maxwell, Q3D, SIwave, PyAEDT); Cadence (custom IC, AWR Microwave Office, Allegro); Calibre, Altium, OrCAD, KLayout

Honors and Certification

Honors: QuestBridge Scholarship; Jack Kent Cooke Foundation Semifinalist; Stanford Intrologic Scholar

Certification: TestOut Network Pro Certification